Semester Project – Machine Learning Project Summaries

Python Machine Learning Workbook for Beginners

The book contains ten machine learning and data science projects. The following summaries focus on the purpose and function of each project, how I can relate it to concepts from C200, and my level of interest.

#	Project	Executive-Level Summary	Connection to C200	Interest
1	**House Price Prediction**	This project predicts the selling price of a house based on information about the house. It learns from existing housing data and then estimates prices for new houses.	I can relate this to C200 because it uses programming concepts such as variables, data, calculations, and using code to solve a real-world problem.	7/10
2	**Spam Email Filtering**	This project determines whether an email message is spam or legitimate. It learns patterns from previously labeled messages and uses those patterns to classify new messages.	This relates to C200 through conditional thinking and working with strings and data. A program is essentially making a decision based on the information it receives.	10/10
3	**Used Car Price Prediction**	This project predicts the selling price of a used car based on information about the vehicle. It learns relationships between vehicle characteristics and their prices.	I can connect this to C200 because the program takes several pieces of input, processes them, and produces an output. It also builds on basic programming and problem-solving skills.	9/10
4	**Stock Market Trend Prediction**	This project uses previous stock-market information to predict future stock trends. The main idea is learning patterns from data that changes over time.	I can relate this to C200 because it involves storing, processing, and analyzing sequences of data. It also shows how programs can use previous information to produce new results.	8/10
5	**Language Translation**	This project creates a system that can take a sentence in one language and produce a translated sentence in another language.	This relates to C200 because text can be treated as data. Concepts such as strings, sequences, inputs, outputs, and program logic are important when processing language.	8/10
6	**Cats and Dogs Image Classification**	This project examines an image and determines whether it contains a cat or a dog. The system learns visual patterns from previously labeled images.	I can relate this to C200 because images are ultimately data that a computer has to process. The project takes an input, processes it, and returns a classification as its output.	8/10
7	**Movie Recommender System**	This project recommends movies based on similarities and relationships between movies. It can use movies a person already likes to suggest other movies they may enjoy.	This reminds me of working with collections of data in C200. A program can compare information, find relationships, and use those results to make a recommendation.	8/10
8	**Face Detection**	This project detects human faces in images and video. It can also identify specific facial areas such as the eyes and smile.	I can relate this to C200 because the program receives data, searches for specific patterns, and produces a result based on what it finds.	9/10
9	**Handwritten English Character Recognition**	This project recognizes handwritten English letters from images. Instead of choosing between only two categories, the system must determine which letter is shown.	This connects to C200 through classification and decision-making. The computer processes an input and determines which of several possible categories it belongs to.	7/10
10	**Customer Segmentation**	This project divides customers into different groups based on their income and spending behavior. Businesses could use these groups to better understand and target different types of customers.	I can relate this to C200 because it involves organizing and analyzing data based on similarities. It shows how programming can turn a large amount of data into useful groups.	7/10

Possible Different Semester Project

Phishing Email Detection Using Machine Learning – This would be a cybersecurity-focused variation of the spam email filtering project. The goal would be to classify email messages and identify messages that may be phishing attempts.